Matrices & Vectors

Section 1: Matrix Inversion & Systems

1 What is the inverse of $A = [[4, 7], [1, 2]]$?

a. $[[2, -7], [-1, 4]]$

b. $[[-2, 7], [1, -4]]$

c. $[[4, -7], [-1, 2]]$

d. $[[2, 1], [7, 4]]$

2 For what value of k does the matrix $[[2, k], [4, 6]]$ fail to be invertible?

a. 3

b. 6

c. 12

d. 0

3 If $AX = B$, where A is invertible, which expression gives the solution column vector X?

a. $A^{-1}B$

b. BA^{-1}

c. B/A

d. AB^{-1}

Section 2: Determinants & Properties

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4 If A and B are 3×3 matrices with $\det(A) = 4$ and $\det(B) = -2$, what is $\det(AB)$?

a. -8

b. 8

c. -2

d. 2

5 If M is a 2×2 matrix with $\det(M) = 6$, what is $\det(M^{-1})$?

a. $\frac{1}{6}$

b. -6

c. 6

d. 36

6 What happens to the determinant of a matrix if two of its rows are interchanged?

a. Its sign is reversed

b. It doubles

c. It remains unchanged

d. It becomes zero

Section 3: Dot Product & Projections

7 Find the angle between $\mathbf{u} = \langle 2, 2 \rangle$ and $\mathbf{v} = \langle 0, 3 \rangle$.

a. 45°

b. 30°

c. 60°

d. 90°

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8 What is the vector projection of $\mathbf{u} = \langle 3, 4 \rangle$ onto $\mathbf{v} = \langle 1, 0 \rangle$?

a. $\langle 3, 0 \rangle$

b. $\langle 4, 0 \rangle$

c. $\langle 0, 4 \rangle$

d. $\langle 1, 0 \rangle$

9 If vectors $\mathbf{a} = \langle 2, x \rangle$ and $\mathbf{b} = \langle 6, -4 \rangle$ are perpendicular, what is x?

a. 3

b. -3

c. 4

d. -4

Section 4: 3D Vectors & Cross Products

10 Compute the cross product $\langle 1, 0, 0 \rangle \times \langle 0, 1, 0 \rangle$.

a. $\langle 0, 0, 1 \rangle$

b. $\langle 0, 0, -1 \rangle$

c. $\langle 1, 1, 0 \rangle$

d. $\mathbf{0}$

11 What is the volume of the parallelepiped determined by vectors $\mathbf{u} = \langle 1, 0, 0 \rangle$, $\mathbf{v} = \langle 0, 2, 0 \rangle$, and $\mathbf{w} = \langle 0, 0, 3 \rangle$?

a. 6

b. 12

c. 3

d. 1

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12 What is the unit vector in the direction of $\mathbf{v} = \langle 3, -6, 2 \rangle$?

a. $\langle \frac{3}{7}, -\frac{6}{7}, \frac{2}{7} \rangle$

b. $\langle \frac{3}{9}, -\frac{6}{9}, \frac{2}{9} \rangle$

c. $\langle 1, -2, \frac{2}{3} \rangle$

d. $\langle \frac{3}{49}, -\frac{6}{49}, \frac{2}{49} \rangle$

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